

04 Face Liveness Detection and Presentation Attack Detection for Registration and Biometric Authentication

Enable the Desktop and Android Registration Clients to process face image streams from biometric devices, perform face liveness and Presentation Attack Detection (PAD), and automatically guide residents, operators, and supervisors through active facial challenges when required to establish that the captured face belongs to a live person.

Complexity Level: High

Overview

The Registration Client shall support **face liveness detection and Presentation Attack Detection (PAD)** for face-based biometric workflows using face image streams received from supported **L0/L1 biometric devices**.

The feature shall be applicable to:

- Resident face capture during registration
- Operator biometric authentication
- Supervisor biometric authentication

The Registration Client shall receive the face frame stream from the connected biometric device and use the configured liveness/PAD capability to determine whether the face presented to the device belongs to a live person and whether a presentation attack is being attempted.

The liveness verification shall follow a **hybrid passive and active approach**.

Initially, the system shall perform **passive liveness detection** using the incoming face frames without requiring the user to perform any specific action. The liveness engine shall generate a quality/confidence score based on the observed facial characteristics and available liveness indicators.

The Registration Client shall compare the passive liveness result against a configurable threshold.

- If the liveness quality meets the configured threshold and no presentation attack is detected, the face capture or authentication workflow shall proceed.
- If the liveness quality does not meet the configured threshold, the system shall automatically initiate **active liveness verification**.

- During active verification, the Registration Client shall prompt the user to perform dynamically selected facial actions.
- The system shall analyse the subsequent frame stream to determine whether the requested action was performed by a live person.
- The face capture or authentication workflow shall proceed only when the configured liveness and PAD criteria are satisfied.

The user shall not select the challenge. The Registration Client shall automatically select and prompt the user with the required action. The challenge sequence should be dynamic and unpredictable.

Potential active challenges may include:

- Blink.
- Smile.
- Turn head left/right.
- Look in a specified direction.
- Other supported facial movements based on the selected liveness technology.

The exact challenge set shall depend on the capabilities of the selected liveness/PAD solution.

The solution shall include **Presentation Attack Detection (PAD)** to protect against applicable attacks such as:

- Printed photographs.
- Digital images displayed on screens.
- Video replay attacks.
- Other supported presentation attack techniques.

The Registration Client shall not accept a face capture or complete biometric authentication when the liveness/PAD assessment fails.

Resident Registration

During resident face capture, the system shall verify liveness before accepting the face biometric.

The workflow shall be:

1. Start face capture.
2. Receive face frames from the L0/L1 device.
3. Perform passive liveness and PAD assessment.
4. Evaluate the liveness quality/confidence.
5. If the configured threshold is satisfied, accept the face capture.

6. If the threshold is not satisfied, automatically initiate an active challenge.
7. Prompt the resident to perform the dynamically selected action.
8. Analyse the subsequent frames.
9. Accept the face only after successful liveness/PAD verification.
10. Allow the resident to retry when verification fails, subject to the configured retry policy.

Operator Authentication

Liveness and PAD shall be performed as part of biometric authentication for operators.

The system shall:

- Receive the operator's face frame stream.
- Perform passive liveness and PAD assessment.
- Initiate active liveness verification when required.
- Prompt the operator to perform the required facial action.
- Validate the action using the incoming frame stream.
- Complete authentication only when the configured liveness/PAD criteria are satisfied.

Where face verification is used for operator authentication, liveness shall provide an additional control to ensure that the matched face is presented by a live person.

Supervisor Authentication

The same liveness and PAD mechanism shall be applied during supervisor biometric authentication.

The system shall ensure that the supervisor is physically present and that the biometric presented during authentication is not a presentation attack.

The system shall automatically initiate an active challenge when passive liveness confidence is insufficient.

Configuration

The face liveness feature shall be configurable at the Registration Client level.

The configuration shall support, at minimum:

- Enable/disable liveness verification.
- Configure passive liveness quality/confidence threshold.

- Configure active liveness enablement.
- Configure the minimum number of active challenges.
- Configure supported challenge types.
- Configure challenge timeout.
- Configure maximum retry count.
- Configure liveness policies for resident, operator, and supervisor workflows.
- Configure behaviour when liveness verification repeatedly fails.

The configuration shall be applicable to both online and offline operation.

Online and Offline Support

The liveness capability shall work in both **online and offline registration environments**.

Core liveness, PAD, active challenge generation, and facial action validation shall be available locally so that network availability does not determine whether liveness security is enforced.

.

The system shall maintain consistent liveness behaviour in:

- Online registration.
- Offline registration.
- Operator authentication.
- Supervisor authentication.

Any configuration or model updates that require connectivity shall be securely synchronized when the Registration Client is online.

Recommended Tech Stack

Desktop

- Java (existing Registration Client technology stack)
- Existing Registration Client device integration framework
- Device SDK/API provided by the biometric device vendor
- Camera/device abstraction layer

Android

- Java/Flutter (existing Android Registration Client architecture)
- Android Camera APIs (where applicable)
- Device SDK/API provided by the biometric device vendor

Integration

- Common Face Capture Device Interface
- Device Adapter Pattern for supporting multiple biometric vendors
- Configurable liveness quality validation
- Existing Registration Client configuration framework

The solution should remain independent of any specific biometric vendor and allow seamless integration with different compliant devices.

Expected Tasks

- Analyse the existing biometric device integration.
- Analyse the existing Desktop and Android face capture workflows.
- Define the face frame streaming interface.
- Define the liveness/PAD processing interface.
- Evaluate suitable liveness and PAD technologies.
- Evaluate the selected solution against applicable ISO/IEC 30107 requirements.
- Implement passive liveness detection.
- Implement Presentation Attack Detection.
- Implement active liveness challenge-response.
- Implement dynamic challenge selection.
- Implement facial action detection.
- Implement configurable liveness thresholds.
- Integrate liveness into resident face capture.
- Integrate liveness into operator authentication.
- Integrate liveness into supervisor authentication.
- Implement online/offline processing.
- Implement retry and failure handling.
- Implement required UI/UX changes.
- Implement configuration changes.
- Implement audit and diagnostic logging.
- Perform performance and interoperability testing.

Mandatory Tasks

1. L0/L1 Device Integration

- Integrate with supported L0/L1 biometric devices.
- Receive face image/frame streams.
- Support continuous or periodic frame processing.
- Handle supported device frame formats.
- Handle device communication failures.
- Handle device connection/disconnection scenarios.
- Provide a common device abstraction for supported devices.

2. Passive Liveness

- Analyse incoming face frames for passive liveness.
- Generate liveness quality/confidence.
- Compare the result against the configured threshold.
- Determine whether active liveness is required.
- Ensure passive liveness works offline.

3. Presentation Attack Detection

- Implement PAD as part of liveness assessment.
- Detect applicable presentation attack types.
- Prevent face capture when PAD detects an attack.
- Prevent biometric authentication when PAD detects an attack.
- Evaluate the implementation against applicable ISO/IEC 30107 requirements.

4. Active Liveness

- Automatically initiate active verification when required.
- Dynamically select the challenge.
- Prompt the user with the selected action.
- Receive and analyse subsequent face frames.
- Verify the requested facial action.
- Re-evaluate liveness during active verification.
- Complete the workflow only after successful verification.

5. Resident Registration

- Integrate liveness into resident face capture.
- Prevent acceptance of a face that does not satisfy liveness requirements.
- Automatically prompt the resident when active verification is required.
- Support retry after unsuccessful verification.
- Enforce the configured retry limit.

6. Operator Authentication

- Integrate liveness into operator biometric authentication.
- Perform passive liveness and PAD.
- Trigger active challenge when required.
- Validate the operator's facial action.
- Prevent authentication when liveness verification fails.
- Ensure liveness works in offline authentication scenarios.

7. Supervisor Authentication

- Integrate liveness into supervisor biometric authentication.
- Perform passive liveness and PAD.
- Trigger active challenge when required.
- Validate the supervisor's facial action.
- Prevent authentication when liveness verification fails.
- Ensure liveness works in offline authentication scenarios.

8. Configuration

Support configuration of:

- Liveness enablement.
- Passive liveness threshold.
- Active liveness policy.
- Challenge types.
- Minimum challenge count.
- Challenge timeout.
- Maximum retry count.
- Workflow-specific liveness policies.

9. Error Handling

Handle:

- Device unavailable.
- Device connection failure.
- Camera unavailable.
- Face not detected.
- Multiple faces detected.
- Poor face quality.
- Liveness quality below threshold.
- PAD failure.
- Active challenge failure.
- Challenge timeout.
- Maximum retries exceeded.
- Invalid/incomplete frame data.
- Liveness engine errors.

10. UI/UX Requirements

1. Face Capture Screen

The face capture screen shall provide:

- Live face preview.
- Face positioning guidance.
- Liveness status.
- Capture progress.
- Clear indication that liveness is being verified.

The UI should provide a simple experience and avoid exposing technical liveness model information unless required for diagnostics.

2. Passive Liveness

While passive liveness is being evaluated, the UI may display:

Checking face liveness...

The system shall automatically proceed when the liveness criteria are satisfied.

The user should not be required to perform any action during passive verification.

3. Active Challenge

When passive liveness does not meet the configured threshold, the Registration Client shall automatically display a challenge.

Examples:

Please blink

Please smile

Please turn your head to the left

The prompt shall be generated by the system and shall not require the user to select an action.



4. Challenge Feedback

The UI should provide real-time guidance while the action is being detected.

Examples:

Please continue

Action detected

Hold still

Good, face captured successfully

5. Liveness Failure

When liveness cannot be established, the UI shall provide clear and actionable guidance.

Example:

We could not verify face liveness. Please try again.

Additional guidance may include:

- Look directly at the camera.
- Ensure your face is clearly visible.
- Improve lighting.
- Keep your face within the capture area.
- Follow the prompted action.

6. PAD Failure

When a potential presentation attack is detected, the UI should avoid exposing detailed security information.

A generic message may be displayed:

Face verification could not be completed. Please try again.

Detailed PAD information should be available only through appropriate diagnostic or audit mechanisms.

7. Retry

The system shall clearly indicate when a retry is available.

If the maximum number of attempts is reached, the UI shall inform the user that the face verification/capture could not be completed and guide them to the applicable recovery process.

8. Operator and Supervisor Authentication

The same interaction principles shall apply during operator and supervisor authentication.

The system shall automatically initiate liveness verification when required and shall not require the operator or supervisor to manually select a liveness test.

Bonus Tasks

- Provide a mock L0/L1 device for development and automated testing.
- Develop automated liveness test scenarios.
- Develop automated PAD attack test scenarios.
- Develop an interoperability testing framework.
- Support liveness model version management.
- Support secure offline model updates.
- Optimize liveness processing for low-resource Android devices.
- Support hardware acceleration where available.

Good-To-Have Tasks

- Support configurable challenge pools.
- Support multiple facial actions.
- Support different liveness policies for resident, operator, and supervisor workflows.
- Provide configurable user guidance.
- Provide diagnostic mode for device and liveness troubleshooting.
- Support multiple biometric device vendors.
- Provide anonymized operational metrics.
- Provide liveness performance metrics such as:
 - Average liveness processing time.
 - Challenge completion time.
 - Liveness retry rate.
 - Liveness failure rate.
- Support device capability discovery.

Deliverables

1. Technical Design

- High-level architecture for integrating L0/L1 biometric devices with the Registration Client.
- Face frame streaming architecture.
- Liveness and PAD processing architecture.
- Passive-to-active liveness decision flow.
- Dynamic challenge mechanism.
- Resident registration integration flow.
- Operator authentication integration flow.
- Supervisor authentication integration flow.
- Online and offline architecture.
- Configuration design.
- Device integration interface.
- Liveness engine interface.
- Error handling and recovery flow.
- Sequence diagrams.
- Security and privacy considerations.
- ISO/IEC 30107 alignment considerations.

2. UI/UX Design

- Updated resident face capture screen.
- Updated operator authentication screen.
- Updated supervisor authentication screen.
- Live face preview.
- Liveness verification status.
- Active challenge prompts.
- Facial action guidance.
- Challenge progress indication.
- Liveness failure messages.
- PAD/failure handling messages.
- Retry flow.
- Maximum retry/failure flow.
- Device error handling.
- Online/offline UI behaviour.
- Consistent Desktop and Android experience.

3. Implementation / Build

- L0/L1 biometric device integration.
- Face frame stream handling.
- Liveness/PAD engine integration.
- Passive liveness detection.
- Active liveness challenge.

- Dynamic challenge selection.
- Facial action validation.
- Configurable liveness threshold.
- Resident face capture integration.
- Operator authentication integration.
- Supervisor authentication integration.
- Retry and failure handling.
- Online/offline support.
- Configuration support.
- Audit and diagnostic logging.
- Desktop Registration Client changes.
- Android Registration Client changes.

Resources

- [ISO/IEC 30107-1:2023](#)
- [ISO/IEC 30107-3:2023](#)
- [ISO/IEC 19794-5:2011](#)
- [Registration Client | MOSIP Docs 1.2.0](#)
- [Android Registration Client | MOSIP Docs 1.2.0](#)